Air Freight Multi-Commodity Routing

MVP Formal Model — Phase 1, v1

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Draft — 2026-05-16

Abstract

Formal specification of the optimization model for the air freight multi-commodity routing problem. The model is a Binary Multi-Commodity Flow problem on a commodity-specific subgraph of the air freight network (see `docs/air_freight_multi_shipment_graph.drawio` and `docs/air_freight_shipment_subgraph.drawio`). Scope: single-shipment routing across multiple flight legs (direct and multi-hop) with two air capacity layers — contracted ULD allocation (Block Space Agreements) and spot rate-card pricing. Probabilistic transit time objectives, through-ULD as a decision variable, period-level utilization commitments, and full bin-packing inside ULDs are deferred. The model parallels the Ocean FCL formulation (`model/ocean_fcl_routing.tex`) for notational consistency.

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1 Problem Statement

Given a set of air shipment requests (commodities), each defined by origin location, destination location, cargo volume, chargeable weight, and time constraints, find a minimum-cost assignment of commodities to paths through the multimodal air freight network such that:

- every commodity is delivered on or before its deadline,
- no flight leg exceeds its ULD physical capacity (volume and weight per ULD),
- no Block Space Agreement (BSA) exceeds its contracted per-flight ULD allocation,
- period-level allocation caps are respected,
- cargo cutoff times are honored,
- hub minimum connect times are honored, including the through-ULD vs. re-ULD distinction, and
- optional per-commodity budget caps are respected.

MVP scope.

- Single-shipment routing; each shipment routes independently subject to shared supply.
- Each scheduled flight leg is one arc. Multi-stop flights (same flight number, intermediate stops) are decomposed into multiple arcs for transit time visibility.
- ULD types: LD3, LD7, PMC pallet, AKE. Optimizer selects ULD type per shipment per flight.
- Two air supply layers per flight: (a) contracted ULD allocation under one or more BSAs; (b) spot rate-card at IATA weight breaks.
- Pivot weight (take-or-pay minimum) modeled on hard BSA contracts.
- Through-ULD vs. re-ULD at hub modeled via pre-computed compatibility parameter ψ, with rehandling cost and MCT differentiated. Explicit decision variable for through-ULD selection deferred to P1.
- Period-level minimum utilization commitments (hard BSA take-or-pay across period) are computed by a separate upstream model and surfaced as fixed per-run targets. This MILP does not solve them jointly.
- Objective: minimize total air + ground freight cost subject to deterministic time-window feasibility. Probabilistic on-time delivery objective deferred to P1.
- Hazmat / perishable / valuable cargo type segregation: model parameter, not a hard MVP constraint (recorded for future restriction logic).

2 Commercial Structure: MAWB and HAWB

Air freight is contracted, billed, and routed at two distinct document layers — Master Air Waybill (MAWB) and House Air Waybill (HAWB). The economic mechanism mid-market forwarders depend on (consolidation of many small HAWBs into one MAWB to access lower tier-break rates) is invisible if a model treats every shipment as an independent tender. This section defines the commercial structure that the formulation must represent.

2.1 Definitions

MAWB *Master Air Waybill.* The contract of carriage between the airline and the freight forwarder. Airline bills the forwarder against one MAWB per shipment from origin airport to destination airport. 11-digit number, 3-digit IATA airline prefix (e.g., `160-12345678` for Cathay). The airline does not see what is inside the MAWB at the pricing level.

HAWB *House Air Waybill.* The contract of carriage between the freight forwarder and the underlying shipper. One HAWB per (shipper, consignee) pair. Issued under the forwarder's own numbering scheme. Airline does not see HAWBs in pricing, but customs at destination does (AMS / ICS2 / equivalent filings are per-HAWB).

Filing mechanics. The MAWB is filed electronically with the airline via IATA Cargo-IMP messages (`FWB` = waybill data, `FHL` = list of HAWBs underneath) or Cargo-XML / ONE Record. The cargo manifest (CARGOMAN) lists all HAWBs under a given MAWB and travels with the physical cargo. At destination, customs receives both the MAWB (carriage record) and each HAWB (as the underlying consignment requiring clearance).

2.2 Universality

The MAWB / HAWB structure applies to *every* air freight shipment regardless of how the forwarder procured capacity. The procurement mode (contracted BSA, spot rate, co-loader / GSA wholesale buy, dynamic NAC quote) determines how the MAWB is *rated*, not whether the MAWB exists. Capacity-side modeling and document-side modeling are orthogonal.

2.3 Direct MAWB (Straight MAWB)

Single large shipment from one shipper to one consignee may be tendered without a separate HAWB: the MAWB serves as both the airline contract and the shipper contract. This is a degenerate case where one MAWB carries exactly one HAWB-equivalent consignment. Common for shipments above ∼ 1,000 kg or for high-value single-piece moves. The MVP formulation must support this as a special case (one HAWB per MAWB), not as a separate construct.

2.4 Co-loader (Neutral Consolidator) Pattern

A forwarder without direct carrier allocation may tender cargo as a HAWB under another forwarder's MAWB (the co-loader holds the carrier paper). From the small forwarder's perspective, the co-loader is a procurement option priced at all-in $/kg with no visibility into the underlying MAWB structure. From the co-loader's perspective, the small forwarder's HAWB is just one of many HAWBs being consolidated. Co-loader procurement is treated in the supply layer as one of the rate-function classes (see Section 5.6, P1 generalization), not as a separate document model.

2.5 Why Consolidation Matters Economically

Airline rate cards apply tier breaks (IATA standard: 45, 100, 300, 500, 1000 kg, plus custom higher breaks on negotiated contracts) at the *MAWB* chargeable weight, not per HAWB. The higher the MAWB weight, the lower the marginal $/kg.

Worked example (TPE→JFK, eight 150 kg shipments).

- Direct tender (each shipment as its own Direct MAWB at +100 tier ∼ $4.80/kg): airline cost $8 \times 150 \times \$4.80 = \$5{,}760$.
- Consol into one MAWB at 1,200 kg total weight at +1000 tier ∼ $2.20/kg: airline cost $1{,}200 \times \$2.20 = \$2{,}640$.

At a forwarder retail HAWB rate of \$4.50/kg (collecting \$5,400 revenue), direct tender yields a \$360 loss while consol yields a \$2,760 margin. The optimization model must be able to find this consolidation; if it cannot, it systematically over-recommends direct tendering and under-quotes the forwarder's margin envelope.

2.6 Constraints That Bind on Consolidation

HAWBs may share one MAWB only if all of the following hold:

(a) Same origin airport $o_{\mathrm{AP}}(k)$ and same destination airport $d_{\mathrm{AP}}(k)$.
(b) Same routing through the air network — every HAWB on the MAWB takes the same flight sequence (i.e., the same set of A_{AIR} arcs).
(c) Cargo-type compatibility — DGR rules at the MAWB level (one DGR HAWB makes the entire MAWB a DGR shipment; carrier and ULD must accept DGR).
(d) Booking-window compatibility — all HAWBs must be ready by the MAWB's CGC.
(e) Aggregate weight / volume fits the ULD set claimed for the MAWB.

2.7 MAWB Chargeable Weight: Density Mixing

Airline billing applies the volumetric rule at the *MAWB* level, not per HAWB. For an MAWB m carrying HAWBs $\{k : h_{k,m} = 1\}$:

$$\mathrm{CW}_m \triangleq \max\left((1+\delta)\cdot\sum_k h_{k,m}\cdot w_k,\ \sum_k h_{k,m}\cdot v_k\cdot 167\right) \tag{1}$$

where δ is the dunnage / strapping factor (typically 3–5%, carrier-specific; default 0.05).

Density mixing benefit. This MAWB-level computation can yield strictly less chargeable weight than summing HAWB-level chargeable weights, because dense cargo absorbs the volumetric "slack" of light cargo. Worked example:

HAWB	Actual wt	Volume	Volumetric (×167)	HAWB CW
H_1 (machinery)	200 kg	0.5 m^3	84 kg	200 kg
H_2 (apparel)	60 kg	1.2 m^3	200 kg	200 kg
Naïve sum of HAWB CW				**400 kg**
MAWB-level ($\delta = 0$)	260 kg	1.7 m^3	284 kg	**284 kg**

The 116 kg saving is captured only by computing CW at the MAWB level. A model that sums HAWB CWs and then applies the tier-break rate cannot see this — it must aggregate physical quantities first.

Modeling note. Eq. (1) is a max of two linear expressions in the binary variables $h_{k,m}$. It linearizes identically to the existing pivot-weight formulation (Section 10, P.10 pattern): introduce $\mathrm{CW}_m \in \mathbb{R}_{\geq 0}$ with two lower-bound inequalities. The linearization is exact because the objective minimizes the rate-tier function applied to CW_m, pushing it down to the binding right-hand side.

2.8 Rate Function on MAWB Chargeable Weight

The airline-billable cost of an MAWB on a flight is a function of the MAWB chargeable weight. The MVP v1 formulation hard-codes two cases: (a) IATA tier-break spot rate (Eq. 10) and (b) pivot-weight BSA. Real procurement spans more shapes:

TACT IATA tiers ($-45, +45, +100, +300, +500, +1000$, plus negotiated $+2000, +3000$). Piecewise-linear with "next-break-down" minimum.

SCR Specific Commodity Rate — discounted lane rate for a 4-digit IATA commodity code. Same piecewise structure, lower rates.

CCR Class Rate — multiplier on general cargo (e.g., VAL +50%, DGR +25%, AVI +75%).

BUC / Pivot
Flat charge per ULD up to a pivot weight, marginal rate above. Generalization of the current P.10 pivot model.

NAC / Block Rate
Flat \$/kg, no tier breaks; all surcharges embedded. Common from co-loaders, GSAs, negotiated lane contracts.

Dynamic Spot
Real-time per-flight quote with rate validity TTL (WebCargo, cargo.one, Awery). Flat \$/kg over the quoted weight band for the TTL window.

V2 generalization: piecewise-linear rate function. The v2 restructure replaces the two hard-coded rate shapes with a single mechanism: each contract c carries a list of (b_i, m_i) tuples — segment breakpoints $b_0 = 0 < b_1 < b_2 < \ldots$ and marginal rates $m_0, m_1, \ldots$ — that define the rate as a piecewise-linear function of MAWB chargeable weight:

$$R_c(\mathrm{CW}) \;=\; \mathrm{floor}_c \;+\; \sum_i m_i \cdot \min\!\big(\max(\mathrm{CW} - b_i, 0),\; b_{i+1} - b_i\big) \tag{2}$$

TACT, SCR, BUC, NAC, BSA-pivot, dynamic spot are all special cases:

Rate type	Encoding as (b_i, m_i) tuples
TACT (TPE-JFK GEN)	$\{(0, 6.00), (45, 5.50), (100, 4.80), (300, 4.20), (500, 3.80), (1000, 3.20)\}$ + IATA next-break-down rule (non-convex check) + min charge floor
BUC (one LD3)	$\{(0, 0), (750, 0.80)\}$ + floor \$1,200
NAC / Block	$\{(0, 3.50)\}$ (single segment)
BSA-pivot tiered	$\{(0, 0), (1000, 0.65), (2000, 0.50)\}$ + floor \$900
Dynamic spot	$\{(0, \text{quoted rate})\}$ + TTL validation

Worked comparison. Same lane, four procurement options, three MAWB chargeable weights:

CW_m	TACT (A)	BUC (B)	NAC (C)	BSA-pivot (D)
200 kg	\$960	\$1,200	\$700	\$900
800 kg	\$3,040	\$1,240	\$2,800	\$900
1,500 kg	\$4,800	\$1,800	\$5,250	\$1,225

At 200 kg NAC wins; at 800 kg and 1500 kg BSA-pivot dominates. The optimizer must choose per-contract, per-MAWB. A single hardcoded cost expression cannot capture this — hence the per-contract piecewise-linear function.

MILP encoding via supply-option assignment. The cleanest formulation does *not* use SOS-2 constraints on a continuous CW. Instead, decompose each contract's rate function into a set of *supply options*, where each option's cost is linear or convex within itself. The global non-convexity of the rate function emerges from a min over options, captured by a single assignment binary per option:

$$\begin{aligned}
& y_{f,m,o} \in \{0,1\} : \text{MAWB } m \text{ on flight } f \text{ uses supply option } o \\
& \sum_{o \in O(c)} y_{f,m,o} = 1 \quad \text{(exactly one option per MAWB-flight)} \\
& \text{cost}_{f,m} = \sum_{o} c_o(\text{CW}_m) \cdot y_{f,m,o}
\end{aligned} \tag{3}$$

Each option o is, e.g., one IATA tier, one BSA pivot segment, one NAC block. Within the option, $c_o(\text{CW}_m)$ is linear (NAC, single TACT tier) or convex (pivot floor + overage = flat then linear ramp). Convex within an option needs no further binaries (LP relaxation is tight on convex PWL minimization).

Why this works: the convexity asymmetry. For minimization of a piecewise-linear function, LP relaxation is tight only when the function is convex (slopes non-decreasing). Concave PWL (slopes non-increasing, e.g., the volume-discount shape of air freight rates) causes the LP to interpolate across the curve via the convex hull, underestimating the true cost. Worked example on a concave PWL with marginal rates $(6.00, 5.50, 4.80)$ on $[0, 45], [45, 100], [100, 300]$:

- True $f(100) = 6 \cdot 45 + 5.5 \cdot 55 = 572.50$.
- LP with continuous λ-formulation, no SOS-2: $\lambda_0 = 2/3$, $\lambda_3 = 1/3$ yields CW $= 100$, cost $= 1532.5/3 = 510.83$. Underestimate of \$61.67 ($\sim 11\%$).

Decomposing into supply options avoids this trap entirely: each option is convex within itself, and the assignment binary $y_{f,m,o}$ selects which option's cost applies. The optimizer naturally picks the cheapest option, which reconstructs the min envelope — equivalent to enforcing SOS-2 on a continuous PWL but operationally cleaner.

Binary count. For TPE-JFK with 6 IATA tiers + BSA-A (2-piece pivot, 2 options) + BSA-B (3-piece tiered, 3 options) + 1 NAC: 12 supply-option binaries per MAWB-flight pair, with one exactly-one constraint. Linear scaling in number of supply options per lane. At MVP scale ($|K| \sim 100$, $|F^k| \sim 30$, ~ 10 options per flight): $\sim 30{,}000$ supply-option binaries — tractable for HiGHS.

This unifies the procurement layer and removes the practitioner-flagged 2015-vs-2026 disconnect. Implementation deferred to v2.

2.9 Volume Measurement

Volume is computed from piece dimensions:

$$v_k = \sum_{p \in \text{pieces}(k)} \frac{L_p \cdot W_p \cdot H_p}{10^6} \ [\text{m}^3]$$

with dimensions in cm and bounding-box convention (smallest enclosing cuboid for irregular pieces). Non-stackable cargo (NSTK flag) incurs a 1.2–1.5× multiplier on v_k, applied at subgraph construction.

2.10 Weight- vs. Volume-Binding ULDs

For typical mid-market mixed cargo ($\sim$ 60% above the 167 kg/m^3 breakeven), the binding constraint on a ULD is determined by ULD geometry, not just cargo:

ULD type	Volume	Weight cap	Volumetric @ 167	Typically binds on...
LD3 / AKE	4.5 m^3	1,587 kg	752 kg	Weight
LD7	11.1 m^3	4,626 kg	1,854 kg	Weight
PMC pallet	7.5 m^3	6,804 kg	1,253 kg	Volume

PMC pallets are weight-over-built relative to volume — cargo cubes out long before weight cap. LD3 / LD7 are balanced toward weight-binding for typical cargo density. This affects ULD selection in the optimization: light/bulky shipments prefer PMC; dense cargo prefers LD3/LD7 to maximize weight per ULD position. Both P.2 (volume) and P.3 (weight) constraints must be retained — the binding one switches by cargo mix.

2.11 Modeling Implication (Formal Restructure in Section 12)

The MVP v1 formulation in this document treats each shipment as a single tender (implicit Direct MAWB per shipment). This degrades the model's representational fidelity for any forwarder that consolidates. The v2 restructure (to be drafted in the next revision) re-roots the decision variables around MAWBs:

- M — set of candidate MAWBs (decision: which MAWBs are created).
- $h_{k,m} \in \{0,1\}$ — HAWB k assigned to MAWB m.
- $y^c_{f,u,m}$ — MAWB m on flight f, ULD type u, contract c.
- Rate tier and pivot weight apply on aggregate MAWB chargeable weight $\mathrm{CW}_m = \sum_k h_{k,m} \cdot cw_k$ (plus dunnage factor), not per HAWB.

This is structurally the LCL pattern from `model/ocean_lcl_routing.tex`: shipments → consol unit → physical transport leg. The v2 draft will mirror the LCL formulation.

3 Graph Structure

3.1 Physical Network Graph $G(N, A)$

The air freight network is a directed graph $G = (N, A)$.

$$N = N_O \cup N_{\mathrm{CFS,O}} \cup N_{\mathrm{AP,O}} \cup N_{\mathrm{AP,H}} \cup N_{\mathrm{AP,D}} \cup N_{\mathrm{CFS,D}} \cup N_D$$

N_O — **Origins** One per shipper origin door (e.g., factory, warehouse). Commodity source nodes.

$N_{\mathbf{CFS,O}}$ — **Origin CFS**
Container Freight Stations near origin airports (e.g., TPE-CFS, KHH-CFS).

ULD build-up location. Airline-owned empty ULDs are delivered here; cargo from one or more shipments is stuffed into a ULD; built ULD is trucked to the airline cargo terminal.

$N_{\mathbf{AP,O}}$ — **Origin airports**
Airline cargo terminals at departure airports (e.g., TPE, KHH). Built ULDs are received here for loading onto aircraft.

$N_{\mathbf{AP,H}}$ — **Hub airports**
Intermediate airports on multi-hop routes (e.g., HKG, NRT, ANC). Through-cargo handling or re-ULDing occurs here.

$N_{\mathbf{AP,D}}$ — **Destination airports**
Airline cargo terminals at arrival airports (e.g., JFK, ORD, LAX). ULDs are unloaded here.

$N_{\mathbf{CFS,D}}$ — **Destination CFS**
CFS near destination airport for ULD breakdown. Trucked from airline cargo terminal; ULDs broken down; individual consignments released for final delivery.

N_D — **Destinations**
One per consignee delivery door. Commodity sink nodes.

$$A = A_{\mathrm{PU}} \cup A_{\mathrm{OC}} \cup A_{\mathrm{AIR}} \cup A_{\mathrm{DC}} \cup A_{\mathrm{FD}}$$

$A_{\mathbf{PU}}$ — **Pickup** (i, j), $i \in N_O$, $j \in N_{\mathrm{CFS,O}}$. Truck pickup from origin door to origin CFS.

$A_{\mathbf{OC}}$ — **Origin CFS to origin airport**
(i, j), $i \in N_{\mathrm{CFS,O}}$, $j \in N_{\mathrm{AP,O}}$. Built ULDs trucked from CFS to airline cargo terminal. Includes fixed CFS build-up time as part of effective arc duration.

$A_{\mathbf{AIR}}$ — **Flight legs**
(i, j), $i, j \in N_{\mathrm{AP,O}} \cup N_{\mathrm{AP,H}} \cup N_{\mathrm{AP,D}}$. One arc per scheduled flight leg (specific carrier, flight number, departure date, origin airport, destination airport). A multi-stop flight (e.g., f: TPE→ANC→JFK) decomposes into two air arcs sharing the same flight number metadata.

$A_{\mathbf{DC}}$ — **Destination airport to destination CFS**
(i, j), $i \in N_{\mathrm{AP,D}}$, $j \in N_{\mathrm{CFS,D}}$. Trucked from airline cargo terminal to CFS. Includes fixed CFS breakdown time.

$A_{\mathbf{FD}}$ — **Final delivery**
(i, j), $i \in N_{\mathrm{CFS,D}}$, $j \in N_D$. Trucked from destination CFS to consignee door.

Remark 1 (Origin and destination CFS as explicit nodes)**.** Unlike ocean (where the POL terminal is one node), air introduces a mandatory CFS step at origin (ULD build-up) and destination (ULD breakdown). Modeling CFS as an explicit node permits: (a) the fixed CFS dwell time to enter feasibility checks, (b) future modeling of CFS dock capacity, ULD availability at CFS, and (c) alternative-CFS routing (e.g., a Kaohsiung shipment trucked to TPE-CFS for consolidation).

Remark 2 (Flight legs vs. flight numbers)**.** A real-world flight number (e.g., 5Y9123 TPE→ANC→JFK) operates one aircraft sortie with an intermediate stop. We model this as *two arcs* in A_{AIR} — $(\mathrm{TPE}, \mathrm{ANC})$ and $(\mathrm{ANC}, \mathrm{JFK})$ — both tagged with the same flight number. This preserves transit time visibility at each leg and provides the structural hook for per-leg ULD allocation accounting. The fact that the ULD physically stays on the aircraft across the intermediate stop is captured by the through-ULD compatibility parameter ψ (Section 5.9).

Figure reference. See the rendered network diagrams: `docs/air_freight_multi_shipment_graph.drawio` (multi-shipment view, supply sharing) and `docs/air_freight_shipment_subgraph.drawio` (single-shipment subgraph $G(N_k, A_k)$ for S3 TPE→NYC). Render the .drawio sources to PDF via drawio export before final approval.

3.2 Commodity-Specific Subgraph $G(N_k, A_k)$

The MILP for commodity k operates on a subgraph $G(N_k, A_k) \subseteq G(N, A)$ containing only arcs that (i) are time-feasible for commodity k, (ii) lie on at least one complete path from $o(k)$ to $d(k)$, and (iii) are compatible with the commodity's cargo type and ULD constraints (e.g., hazmat-restricted flights excluded for non-DGR-certified carriers). See Section 6 for the construction algorithm.

4 Sets and Indices

Symbol	Description
K	Set of commodities (shipments), indexed by k
N, A	All nodes and arcs
A_{PU}, A_{OC}, A_{AIR}, A_{DC}, A_{FD}	Arc subsets by type
$A^k \subseteq A$	Feasible arc set for commodity k (Section 6)
$A^k_S \triangleq A^k \cap A_S$	Feasible arcs of type S for commodity k
N_k	Nodes incident to A^k
F	Set of scheduled flight legs, indexed by f. Each flight is one arc in A_{AIR}.
$F^k \subseteq F$	Flights corresponding to arcs in A^k_{AIR}
$f(i, j)$	Mapping: air arc $(i, j) \in A_{\text{AIR}}$ to its flight $f \in F$
U	Set of ULD types: $U = \{\text{LD3, LD7, PMC, AKE}\}$
$U_f \subseteq U$	ULD types available on flight f (depends on aircraft type)
C	Set of ULD allocation contracts (BSAs), indexed by c
$C_f \subseteq C$	Contracts that allocate ULDs on flight f
T	Set of allocation time periods (e.g., monthly), indexed by t
$\tau_k(i)$	Earliest arrival time of commodity k at node i
$\text{Hub}(j)$	Set of consecutive air-arc pairs at hub node $j \in N_{\text{AP,H}}$ for shipment k: $\{((i, j), (j, \ell)) : (i, j), (j, \ell) \in A^k_{\text{AIR}}\}$

5 Parameters

5.1 Commodity Parameters

Parameter	Symbol	Description	Unit
Volume	v_k	Cargo volume (sum of all pieces)	CBM
Actual weight	w_k	Gross weight	kg
Chargeable weight	cw_k	$\max(w_k,\ v_k \cdot 167)$	kg
Piece dimensions	D_k	Max single-piece L×W×H (for ULD fit check)	cm
Cargo ready	t_k^{rdy}	Earliest pickup time from origin door	datetime
Deadline	T_k^{dead}	Latest acceptable arrival at destination	datetime
Origin	$o(k)$	Source node $\in N_O$	—
Destination	$d(k)$	Sink node $\in N_D$	—
Cargo type	τ_k	General / DGR / PER / VAL / HEA	categorical
HS code	hs_k	6-digit Harmonized System code	—
Incoterm	inco_k	EXW / FCA / CIF / DDP	—
Budget cap	B_k	Hard cost ceiling; ∞ if not set	USD

Chargeable weight.

$$cw_k \triangleq \max(w_k,\ v_k \cdot 167) \tag{4}$$

IATA standard: $1\,\mathrm{m}^3 = 167\,\mathrm{kg}$ volumetric equivalent. Dense cargo (machinery, electronics) rated on w_k; bulky-light cargo (furniture, packaging) rated on volumetric.

5.2 Ground Arc Parameters (Pickup, CFS, Final Delivery)

For each ground arc $(i, j) \in A_{\mathrm{PU}} \cup A_{\mathrm{OC}} \cup A_{\mathrm{DC}} \cup A_{\mathrm{FD}}$:

Parameter	Symbol	Description	Unit
Cost	c_{ij}^{G}	Truck cost per shipment	USD
Mean transit	μ_{ij}^{G}	Mean transit time	days
Std dev	σ_{ij}^{G}	Std dev of transit	days
CFS build-up time	β^{O}	Fixed origin CFS dwell (ULD build-up). Configurable.	hours
CFS breakdown time	β^{D}	Fixed destination CFS dwell (ULD breakdown). Configurable.	hours

The effective transit time on the CFS→airport arc $(i, j) \in A_{\mathrm{OC}}$ includes the CFS build-up: $\tilde{\mu}_{ij}^{\mathrm{OC}} = \mu_{ij}^{\mathrm{OC}} + \beta^{\mathrm{O}}$. Likewise for A_{DC} with β^{D}.

5.3 Flight (Air Arc) Parameters

For each flight $f \in F$ (equivalently, each $(i, j) \in A_{\mathrm{AIR}}$):

Parameter	Symbol	Description	Unit
Operating carrier	$\text{carrier}^{\text{op}}(f)$	IATA code of the carrier physically operating the aircraft. Determines ULD pool ownership, ψ pre-computation (Section 5.9), and contract / capacity allocation.	—
Marketing carrier	$\text{carrier}^{\text{mk}}(f)$	IATA code of the carrier issuing the AWB / marketing the flight; may differ on codeshares. Used for billing presentation only; operational eligibility keys on $\text{carrier}^{\text{op}}$.	—
Flight number	$\text{fno}(f)$	Flight number; multiple legs may share a number	—
Aircraft type	$\text{ac}(f)$	777F, 747-8F, A330F, 777-300ER, A350-900, ...	—
ETD	ETD_f	Scheduled departure datetime	datetime
ETA	ETA_f	Scheduled arrival datetime	datetime
Block time	μ_f^{AIR}	$\text{ETA}_f - \text{ETD}_f$	hours
Std dev	σ_f^{AIR}	Std dev of actual block time	hours
Cargo cutoff	CGC_f	Latest time ULDs must be tendered to airline cargo terminal	datetime
Document cutoff	DCO_f	Latest time MAWB+HAWB filed with airline (Cargo-IMP `FWB`/`FHL`, Cargo-XML, ONE Record)	datetime
US pre-load filing	AMS_f	Pre-load CBP filing deadline (US destinations only)	datetime
EU pre-load filing	ICS2_f	Pre-load ENS filing deadline (EU destinations only)	datetime
Canada pre-load filing	ACI_f	Pre-load CBSA filing deadline (Canada destinations only)	datetime
DGR certified	$\text{dgr}(f)$	Accepts dangerous goods?	boolean
Cold chain	$\text{per}(f)$	Maintains $\leq 8°\text{C}$?	boolean

Cutoff set and effective cutoff. Define the set of applicable cutoffs for flight f:

$$\text{CO}_f \triangleq \{\,\text{CGC}_f,\ \text{DCO}_f\,\} \cup \begin{cases} \{\text{AMS}_f\} & \text{if } d(k) \in \text{US destinations} \\ \{\text{ICS2}_f\} & \text{if } d(k) \in \text{EU destinations} \\ \{\text{ACI}_f\} & \text{if } d(k) \in \text{Canada destinations} \\ \emptyset & \text{otherwise} \end{cases} \tag{5}$$

Cargo and document preparation must complete before the *most restrictive* of these. The effective cutoff is the minimum over the set:

$$\text{CO}_f^* \triangleq \min_{c \in \text{CO}_f} c \tag{6}$$

Typical timing (offsets from ETD_f): $\text{CGC}_f = \text{ETD}-4\text{h}$ to -8h (cargo-type dependent), $\text{DCO}_f = \text{ETD}-4\text{h}$ to -6h, $\text{AMS}_f = \text{ETD}-4\text{h}$ to -5h, $\text{ICS2}_f = \text{ETD}-4\text{h}$, $\text{ACI}_f = \text{ETD}-4\text{h}$. Equivalently, the lead time required is $\text{LT}_f^* = \max_c(\text{ETD}_f - c)$ and $\text{CO}_f^* = \text{ETD}_f - \text{LT}_f^*$.

Per-shipment document preparation time. Filing the MAWB+HAWB stack is not instant; it depends on shipment complexity:

$$\text{prep\_time}_k \triangleq \tau^{\text{base}} + \alpha \cdot |\text{HAWBs}_k| + \mathbb{1}[\tau_k = \text{DGR}] \cdot \tau^{\text{DGR}} + \mathbb{1}[\tau_k = \text{PER}] \cdot \tau^{\text{PER}} + \tau^{\text{customs}}(k) \tag{7}$$

Defaults: $\tau^{\text{base}} = 30$ min, $\alpha = 1.5$ min per HAWB, $\tau^{\text{DGR}} = 3$ h, $\tau^{\text{PER}} = 1$ h, τ^{customs} 0–2 h based on customs complexity. All parameterizable per tenant.

The binding feasibility check at the origin airport node $a \in N_{\text{AP,O}}$ is:

$$\tau_k(a) + \text{prep\_time}_k \;\leq\; \text{CO}^*_f \tag{8}$$

This replaces the single-cutoff check on CGC_f in the subgraph construction (Section 6) and in constraint P.11.

5.4 ULD Specifications

IATA ULD code convention. ULD codes are 3-letter identifiers with structural meaning:

- **1st letter**: family. `P` = Pallet, `A` = Aircraft container (rigid walls), `R` = Reefer, `U` = Non-certified.
- **2nd letter**: base dimensions. `M` = 96"×125" (main-deck size), `K` = 60.4"×61.5", `L` = 60.4"×125".
- **3rd letter**: contour / shape. `C` = standard cube, `E` = contoured, `P` = lower-deck contour.

So **PMC** = Pallet, 96"×125" base, cube contour: a flat aluminum pallet 244 × 318 cm with cargo stacked on top and secured with a strapping net. PMCs are main-deck freighter ULDs. Contrast with **LD3 (AKE/AVE)**: a rigid-walled container used in wide-body belly compartments — cargo is stuffed inside through a door. The distinction matters for loadability (pallets have more volumetric loss to net gaps and contour) and aircraft compatibility (PMCs only fit freighters; LD3s fit both passenger belly and freighter).

ULD type	Code	Family	Volume	Payload	Aircraft compatibility
LD3	AKE/AVE	Container	4.5 m^3	1,587 kg	Wide-body belly + freighter
LD7	ALE/AAY	Container	11.1 m^3	4,626 kg	Wide-body belly + freighter
PMC	PMC/PAG	Pallet	7.5 m^3	6,804 kg	Freighter main deck only
AKE small	AKE	Container	4.5 m^3	1,497 kg	Wide-body belly

For each $u \in U$:

Parameter	Symbol	Description	Unit
Volume capacity	V_u	ULD usable volume (Table above)	CBM
Weight capacity	W_u	ULD payload limit	kg
Fill rate cap	η	Max fraction of CBM usable in practice (default 0.97)	—

For each flight f and ULD type $u \in U_f$:

Parameter	Symbol	Description	Unit
Aircraft ULD positions	$P_{f,u}$	Number of u-type ULD positions on $\text{ac}(f)$	—

$P_{f,u}$ is an upper bound on physical positions for ULD type u on the aircraft. It is independent of the forwarder's contracted allocation.

5.5 Procurement Types and Supply Layer Catalog

The forwarder procures air capacity through several structurally distinct channels. The rate-function generalization (Section 2.8) handles *pricing* differences across channels via per-contract piecewise-linear functions. This section catalogs the *structural* differences: who holds the allocation, who issues the MAWB, how allocations are accounted, and whether the procurement decision is at the MAWB or HAWB level.

Supply type enum. Each contract $c \in C$ carries a supply_type(c) attribute:

$$\text{supply\_type}(c) \in \{\texttt{DIRECT\_BSA\_HARD}, \texttt{DIRECT\_BSA\_SOFT}, \texttt{GSA}, \texttt{COLOADER}, \texttt{DYNAMIC\_SPOT}\}$$

Supply types catalog.

Type	Allocation held by	Tenant contract with	MAWB issued by	Rate function shape	Allocation accounting
`DIRECT_BSA_HARD`	Tenant	Carrier	Tenant	Pivot + tiered overage; $\underline{M}_{c,u,t} > 0$	Per-flight + per-period + take-or-pay
`DIRECT_BSA_SOFT`	Tenant	Carrier	Tenant	Tiered (BUC or TACT); $\underline{M}_{c,u,t} = 0$	Per-flight, no take-or-pay
`GSA`	GSA	GSA	Tenant (under GSA's master)	Marked-up tier or all-in	Per-flight (GSA's pool)
`COLOADER`	Co-loader	Co-loader	**Co-loader, not tenant**	All-in NAC per quote	Per-shipment quote, no advance allocation
`DYNAMIC_SPOT`	None	Airline / platform	Tenant	Single-segment quote with TTL	None

Mapping to existing parameter blocks.

- Rate function $R_c(\text{CW}_m)$ (Section 2.8): applies to all types. Each supply type populates the (b_i, m_i) tuples differently.
- Per-flight allocation alloc(c, f, u) and period allocation alloc(c, u, t) (Section 5.6): apply to `DIRECT_BSA_HARD`, `DIRECT_BSA_SOFT`, `GSA`. For `COLOADER` and `DYNAMIC_SPOT`: not applicable (alloc $= \infty$ effective).
- Take-or-pay minimum $\underline{M}_{c,u,t}$ (Section 5.6): nonzero only for `DIRECT_BSA_HARD`.
- Pivot weight $\pi_{c,u}$ (Section 5.6): only meaningful for `DIRECT_BSA_HARD` and `DIRECT_BSA_SOFT` when the contract rates per-ULD.
- Rate validity expiry$_c$: only meaningful for `DYNAMIC_SPOT`; ignored elsewhere.

Co-loader as structural dual-mode. `COLOADER` procurement is the only type where the tenant does *not* issue its own MAWB — instead, the tenant hands HAWBs to a co-loader (a larger forwarder, e.g., K+N, DSV, Geodis) who consolidates them under their own MAWB. The tenant is effectively a HAWB-shipper on someone else's MAWB. This affects the v2 MAWB-centric formulation: tenant MAWB-level decision variables $y^c_{f,u,m}$ (Section 2) do not apply to co-loader-routed HAWBs. Instead, introduce a HAWB-level binary:

$$\text{coloader}_{k,o} \in \{0,1\} : \text{HAWB } k \text{ uses co-loader option } o \in O_{\text{COLOADER}} \tag{9}$$

with exactly-one constraint across procurement paths per HAWB:

$$\sum_{m \in M} h_{k,m} + \sum_{o \in O_{\text{COLOADER}}} \text{coloader}_{k,o} = 1$$

The co-loader handles all downstream routing, ULD selection, and consolidation opaque to the model. Cost: lose density-mixing visibility into co-loader's consolidation, but gain a procurement path that mid-market forwarders without direct carrier paper rely on heavily. Full formalization is part of the v2 MAWB restructure (Section 12).

GSA as marked-up direct contract. `GSA` contracts are structurally identical to direct BSAs: forwarder issues MAWB, GSA holds the allocation, MILP machinery applies unchanged. The only difference is the rate function shape — GSAs typically add a 3–8% markup on the underlying carrier rate, encoded directly in the (b_i, m_i) tuples. No new constraints or variables; `GSA` is a tag on the existing contract entity.

Dynamic spot as single-segment with TTL. `DYNAMIC_SPOT` contracts have:

- Rate function: single segment $\{(0, \text{quoted\_rate})\}$ + optional minimum charge.
- No allocation ($\text{alloc} = \infty$).
- Rate validity attribute expiry_c: planning horizon must end before expiry_c (validated at the data layer).

Capacity availability uncertainty (RQ / KK / NS booking responses) is a separate scope decision (`booking confirmation` modeling; deferred to a later point).

Out of MVP scope.

- **Charter capacity** — full-aircraft buyout, edge case for mid-market forwarders; defer.
- **Broker-of-record / freight-forwarder-as-customer** — multi-level forwarding hierarchies (forwarder buying from another forwarder buying from a co-loader); defer, model treats each tier as a separate co-loader relationship.
- **Alliance slot sharing** (`SkyTeam`, `Star Alliance`, `oneworld`): cross-carrier capacity exchange; already deferred in Section 12.

5.6 Contracted ULD Allocation (Block Space Agreement) Parameters

A Block Space Agreement (BSA) reserves a fixed allocation of ULD positions per flight at a contracted rate. For each contract $c \in C$:

Parameter	Symbol	Description	Unit
Flights covered	$F_c \subseteq F$	Set of scheduled flights covered by contract c	—
Per-flight allocation	$\mathrm{alloc}(c, f, u)$	ULD count of type u allocated on flight f under contract c	ULDs
Period allocation	$\mathrm{alloc}(c, u, t)$	Total ULDs of type u allocated in period t under contract c	ULDs
Period utilization	$\mathrm{util}(c, u, t)$	Already-booked ULDs in period t at solve time	ULDs
Remaining (period)	$\mathrm{rem}(c, u, t)$	$\mathrm{alloc}(c, u, t) - \mathrm{util}(c, u, t)$	ULDs
Min utilization (period)	$\underline{M}_{c,u,t}$	Hard BSA minimum: external upstream model output. Default 0 for soft BSA.	ULDs
Pivot weight	$\pi_{c,u}$	Minimum chargeable weight per ULD per flight under c	kg/ULD
Contract rate	r_c	Contracted price	USD/kg chargeable
Hard vs. soft	hard_c	Boolean: hard BSA (take-or-pay) or soft (cancellable)	—

Remark 3 (Period commitments are upstream)**.** $\underline{M}_{c,u,t}$ is computed by a separate upstream model that allocates how much of each hard BSA's monthly take-or-pay commitment should be met by each routing run within the period. The routing MILP takes $\underline{M}_{c,u,t}$ as a fixed input per run. The MILP does *not* simultaneously optimize period-level commitment allocation; that is a higher-level decision.

Remark 4 (Pivot weight semantics)**.** The pivot $\pi_{c,u}$ is the minimum chargeable weight per ULD per flight regardless of actual cargo loaded. If a forwarder claims one LD3 with pivot 1,000 kg on a flight but loads only 400 kg of cargo, the forwarder pays for $1{,}000\,\mathrm{kg} \cdot r_c$. This is the take-or-pay floor on hard BSAs.

5.7 Spot Rate-Card Parameters

For each flight f, the airline offers spot rate-card pricing with IATA weight breaks:

Parameter	Symbol	Description	Unit
Break tiers	$B = \{N, 45, 100, 300, 500, 1000\}$	IATA standard break points (kg)	kg
Tier rate	$r^{\mathrm{sp}}_{f,b}$	Rate per kg in tier b on flight f	USD/kg
Tier minimum	$b_{\min}$	N tier: minimum charge floor	USD

Spot rate cost per shipment. For shipment k tendered on flight f at the spot rate, the cost is the minimum across applicable tiers:

$$\mathrm{cost}^{\mathrm{spot}}(k, f) \triangleq \min_{b \in B} r^{\mathrm{sp}}_{f,b} \cdot \max(cw_k,\, b) \tag{10}$$

Since cw_k is a parameter (known at planning time), $\mathrm{cost}^{\mathrm{spot}}(k, f)$ is pre-computable. The IATA rule that "charging at the next tier with a lower rate may yield a lower total cost" is captured automatically by the min.

5.8 Surcharge Parameters

Per shipment per flight, the surcharge stack adds:

Parameter	Symbol	Description	Unit
Fuel surcharge	FSC_f	USD/kg, monthly refresh	USD/kg
Security surcharge	SSC_f	USD/kg	USD/kg
Origin THC	THC_f^O	Per shipment, origin terminal handling	USD
Destination THC	THC_f^D	Per shipment, destination terminal handling	USD
AMS fee (US imports)	AMS_f	Per shipment	USD

5.9 Hub Connection Parameters (Through-ULD vs. Re-ULDing)

For each pair of consecutive air arcs (f, g) at hub j — meaning f ends at j and g starts at j — and each ULD type u:

Parameter	Symbol	Description	Unit
Through-ULD compatible	$\psi_{f,g,u}$	1 if a ULD of type u can transit hub j without being broken down (same aircraft for through-flights; or same airline through-cargo agreement)	boolean
MCT (through)	$\mathrm{MCT}_{f,g}^{\mathrm{thru}}$	Min connect time when through-ULD is used (technical-stop turnaround or hub through-cargo time)	hours
MCT (re-ULD)	$\mathrm{MCT}_{f,g}^{\mathrm{reULD}}$	Min connect time when ULD must be broken down and rebuilt at hub	hours
Rehandling cost (same-ULD)	$\rho_{f,g,u}^{\mathrm{reULD}}$	Cost per ULD when same type u is used on both legs but $\psi_{f,g,u} = 0$ (e.g., interline pair without ULD interchange agreement)	USD/ULD
Rehandling cost (cross-ULD)	$\bar{\rho}_{f,g}^{\mathrm{reULD}}$	Cost when different ULD types are used on f and g (e.g., LD3 belly to PMC freighter); typically equal to or slightly higher than ρ^{reULD} since cargo restuffs into a different shape	USD/ULD

Definition 1 (Re-ULDing)**.** *Re-ULDing* is the operation of breaking down a built ULD at an intermediate (hub) airport and stuffing its cargo into a different ULD for the onward flight. It occurs when:

(a) the two consecutive flights are operated by *different airlines* (interline routing) — each airline owns a distinct ULD pool and ULDs cannot transfer;
(b) the two flights require *different ULD types* (e.g., LD3 on a passenger belly leg and PMC main-deck pallet on a freighter leg);
(c) the hub lacks through-cargo facilities for this ULD type (rare at major hubs).

Re-ULDing is operationally expensive: it adds $\sim$ 4–8 hours of dwell time (breakdown + storage + rebuild), incurs ~\$150–\$500 in handling fees per ULD, and introduces cargo damage risk. When $\psi_{f,g,u} = 1$, the ULD physically stays intact across the hub — either remaining on the same aircraft (through-flight technical stop) or moving as a complete unit between aircraft at hub through-cargo facilities.

ULD interchange set. Let $\Pi \subseteq \{ (c_1, c_2, u) \}$ be the *ULD interchange set*: triples for which carriers c_1 and c_2 have a bilateral or alliance-level IATA ULD-CPM (Container Pallet Movement) agreement covering ULD type u. When $(c_1, c_2, u) \in \Pi$, a built ULD of type u can physically transfer from c_1's pool to c_2's pool at a hub with cross-airline transfer facilities — no breakdown, no rebuild, no fee. Typical sources for Π:

- Star Alliance ULD pool: AC, LH, NH, OZ, SK, UA, ... covering common types (LD3, PMC, AKE).
- SkyTeam ULD pool: AF, KL, DL, KE, MU, ...
- oneworld ULD pool: AA, BA, CX, JL, QF, ...
- Bilateral non-alliance agreements (limited; e.g., cargo-only carrier pairs).

For MVP synthetic instances, Π is populated from a static table of alliance memberships intersected with the ULD type catalog. For production, the source is each airline's published ULD interchange directory or the IATA ULD-CPM database.

MVP rule for pre-computing $\psi_{f,g,u}$. $\psi_{f,g,u}$ is a *pre-computed parameter* populated before solve. The rule keys on the *operating* carrier $\text{carrier}^{\text{op}}(\cdot)$, not the marketing carrier — ULD pool ownership follows the airline that physically operates the aircraft, not the airline whose code appears on the AWB. Set $\psi_{f,g,u} = 1$ if and only if $u \in U_f \cap U_g$ *and* at least one of the following holds:

(a) **Through-flight (technical stop).** $\text{fno}(f) = \text{fno}(g)$ *and* $\text{carrier}^{\text{op}}(f) = \text{carrier}^{\text{op}}(g)$. The ULD never leaves the aircraft.
(b) **Same-airline through-cargo at hub.** $\text{carrier}^{\text{op}}(f) = \text{carrier}^{\text{op}}(g)$ *and* the hub j supports through-cargo handling for type u at this carrier's terminal.
(c) **Interline ULD interchange.** $(\text{carrier}^{\text{op}}(f), \text{carrier}^{\text{op}}(g), u) \in \Pi$ *and* the hub j supports cross-airline ULD transfer for the relevant pool.

Otherwise $\psi_{f,g,u} = 0$ and re-ULDing is required.

Remark 5 (Why ψ is a parameter, not a decision variable — and why this is the right MVP choice)**.** Whether a built ULD can ride through a hub is determined by physical and contractual realities that exist before the forwarder plans the move:

- Which carrier operates each leg (codeshares are irrelevant — only the operator's ULD pool matters).
- Whether the hub has the through-cargo handling infrastructure for that ULD type.
- Whether a bilateral or alliance ULD-CPM agreement exists between the operators for that ULD type.

The forwarder cannot "elect" through-ULD where the operational reality doesn't allow it; conversely, the forwarder generally has no operational reason to elect re-ULDing when through-ULD is available — it just adds dwell time and rehandling fees with no benefit. The single edge case where the forwarder *could* elect re-ULD despite $\psi = 1$ is hub re-sorting (splitting one ULD's cargo across multiple onward flights). That edge case is rare in the mid-market forwarder context, structurally distinct, and deferred to P1; see Section 12, item 2. **Bottom line for MVP:** ψ is a data input. The model only needs the right data; no decision-variable promotion is needed. If you (future-reader) are considering revisiting this, the question is not "should ψ be a decision variable?" — the question is "does the routing-with-resort use case justify the model complexity?" For MVP, no.

Reading the rule with an example. Consider shipment k routing TPE $\rightarrow$ JFK across three candidate hub connections, with u = PMC pallet:

- *Path A — CX564 TPE-HKG + CX880 HKG-JFK (same airline).* Rule (b) applies: $\text{carrier}^{\text{op}} = \text{CX}$ on both legs; HKG has Cathay through-cargo for PMC. $\psi_{\text{CX564, CX880, PMC}} = 1$. Hub MCT $\approx$ 3h, no rehandling fee.
- *Path B — AC2 TPE-YYZ + LH471 YYZ-JFK (Star Alliance interline).* Rule (c) applies: $(\text{AC}, \text{LH}, \text{PMC}) \in \Pi$ via Star Alliance pool; YYZ supports cross-airline transfer. $\psi = 1$. Hub MCT $\approx$ 3–4h, no rehandling fee. (A naïve "same-carrier-only" rule would incorrectly return $\psi = 0$ here and over-count re-ULDing on alliance interline routings — this is the bug that motivated Task #6's correction.)
- *Path C — BR32 TPE-LAX + AA90 LAX-JFK (non-alliance interline).* Neither (b) nor (c) applies: BR is Star Alliance, AA is oneworld; no joint ULD pool. $\psi = 0$. Re-ULDing required at LAX: $\sim$6–12h dwell, $\sim$\$150–500/ULD fee.

The MILP uses ψ in constraint P.14 (Section 8) to select the applicable MCT, and in the objective (Section 9) to attribute rehandling cost.

5.10 Embargo Parameters

Embargoes are carrier-, route-, commodity-, and time-bounded prohibitions on cargo acceptance. They are imposed by carriers (operational policy, security incidents, peak capacity), by regulators (geopolitical, sanctions), or by IATA / industry notice. The model represents embargoes as a parameter set used at subgraph construction, not as MILP constraints — the optimizer never sees embargoed arcs.

Embargo record schema. Let E be the set of embargo records. Each $e \in E$ is a tuple:

Field	Description
carrier(e)	IATA carrier code, or `*` wildcard
origin(e)	Origin airport code, region prefix (e.g., `CN-*`), or `*`
dest(e)	Destination airport code, region prefix, or `*`
hub(e)	Transit hub airport code, or $\emptyset$ (applies regardless of routing)
ac_type(e)	`FREIGHTER` / `PAX_BELLY` / `*`
cargo_type(e)	`GEN` / `DGR` / `PER` / `VAL` / `AVI` / `HUM` / `*`
commodity(e)	4-digit IATA commodity code, or `*`
dgr_class(e)	DGR class+subclass (e.g., `9-II`), or `*`
lithium_pi(e)	PI965 / PI966 / PI967 / PI968 / PI969 / PI970 / $\emptyset$ (see Section 5.11)
start(e), end(e)	Date-time validity range
source(e)	`carrier_notice` / `iata` / `regulatory` / `tenant_internal`

Active embargoes for a flight.

$$E_f \triangleq \{e \in E : \text{start}(e) \leq \text{ETD}_f \leq \text{end}(e)\} \tag{11}$$

Match predicate. Embargo record e matches the (commodity k, flight f) pair, denoted $\text{match}(e, k, f)$, iff every non-wildcard field of e matches the corresponding attribute of the shipment-flight pair:

- carrier and aircraft type from f,
- origin and destination airports from $(o_{\text{AP}}(k), d_{\text{AP}}(k))$ or from arc endpoints,
- transit hub from the path through which (k, f) connects (or $\emptyset$ for any-routing records),
- cargo type / commodity / DGR class / lithium PI from k's commodity attributes.

Wildcard fields in e match anything. Records in E_f OR together: a flight-arc is embargoed for k iff *any* record in E_f matches.

Arc embargo-feasibility.

$$\text{embargo\_ok}(k, f) \triangleq \neg \exists\, e \in E_f : \text{match}(e, k, f) \tag{12}$$

Examples (illustrative records).

Type	Record
Carrier-commodity	carrier = `CX`, origin = `TPE`, dest = `US-*`, hub = `HKG`, lithium_pi = `PI965`, start = 2023-06-01, end = 2099-12-31
Seasonal Hajj	carrier = `EK`, dest = `JED`, cargo_type = `GEN`, start = 2026-06-10, end = 2026-07-22
CNY perishables	origin = `HKG`, cargo_type = `PER`, start = 2026-02-08, end = 2026-02-22
Aircraft-type DGR	carrier = `*`, ac_type = `PAX_BELLY`, lithium_pi = `PI965`, start = 2024-04-01, end = 2099-12-31

Sourcing. For MVP, E is a manually-curated table maintained by tenant ops, seeded from carrier embargo portal pages, IATA alerts (TACT supplements), and operational knowledge. P1: agent capability to parse carrier notices (email, Slack) into structured records with human review. P2: WebCargo / cargo.one embargo API if exposed.

Scope decisions for MVP.

- All embargoes are hard prohibitions. Soft / capacity-driven restrictions are handled in the rate / capacity layer (Section 5.6), not as embargoes.
- Individual carriers only; alliance grouping (`SkyTeam`, `Star Alliance`, `oneworld`) deferred to P1.
- Embargo set is global per tenant (carrier-imposed embargoes affect all shippers). Tenant-specific carrier exclusions are handled separately as carrier blacklist (Section 12, distinct from embargoes).
- Lithium PI codes referenced by lithium_pi(e) are resolved by the lithium taxonomy in the next revision (separate scope decision).

5.11 Lithium Battery Taxonomy and Per-Flight Acceptance

Lithium batteries account for an estimated 30–40% of TPE/HKG eastbound air cargo in 2026. They are not a single commodity class: the IATA Dangerous Goods Regulations distinguish six packing instructions and three section tiers per PI, and each carrier publishes per-route acceptance rules that vary by PI, section, and aircraft type. Modeling lithium as a generic "DGR" boolean conflates flights that accept PI967-contained-in-equipment with those that refuse PI965 IA standalone cells.

IATA lithium taxonomy.

UN #	Chemistry	Configuration	PI	
UN3480	Li-ion	Standalone	PI965	
UN3481	Li-ion	Packed with equipment	PI966	
UN3481	Li-ion	Contained in equipment	PI967	
UN3090	Li-metal	Standalone	PI968	(forbidden on passenger aircraft globally since 20
UN3091	Li-metal	Packed with equipment	PI969	
UN3091	Li-metal	Contained in equipment	PI970	

Within each PI, the IATA *Section* tier determines regulatory burden:

Section IA
: Large cells / batteries ($>$ 100 Wh Li-ion, $>$ 2 g Li metal). Fully regulated. Most restricted.

Section IB
: $\leq$ 100 Wh Li-ion / $\leq$ 2 g Li metal in larger packaging. Partial relief.

Section II
: Small batteries below threshold quantities. Most operational relief, still regulated.

Commodity lithium attributes. Extend the commodity parameter block with an optional lithium_spec_k record ($\emptyset$ for non-lithium shipments):

Field	Description
un_number	`UN3480` / `UN3481` / `UN3090` / `UN3091`
pi_code	`PI965` / `PI966` / `PI967` / `PI968` / `PI969` / `PI970`
section	`IA` / `IB` / `II`
watt_hours	Maximum Wh per battery in the shipment
soc_compliant	Boolean: shipper-certified SOC $\leq$ 30% of rated capacity (required for UN3480, UN3090 standalone)
ddr	Boolean: damaged / defective / recalled. Universally refused in MVP.

Per-flight acceptance matrix. For each flight f with aircraft type indicator $\text{ac\_type}(f) \in \{\texttt{FREIGHTER}, \texttt{PAX\_BELLY}\}$, the carrier publishes an acceptance matrix:

$$\text{lithium\_accept}_f : (\text{PI}, \text{Section}, \text{ac\_type}) \to \{\text{True}, \text{False}\} \tag{13}$$

Whitelist semantics: default is `False` (refuse). A flight accepts a (PI, Section) combination only if the matrix explicitly says `True`. The cost asymmetry (false-accept implies 48 h roll, fire-risk liability) justifies a default-deny posture.

Lithium feasibility predicate. For shipment k with $\text{lithium\_spec}_k \neq \emptyset$ and flight f:

$$\begin{aligned}\text{lithium\_ok}(k, f) \triangleq\ & \neg\, \text{lithium\_spec}_k.\text{ddr} \\ & \wedge \big(\text{lithium\_spec}_k.\text{un\_number} \in \{\texttt{UN3480}, \texttt{UN3090}\} \;\Rightarrow\; \text{lithium\_spec}_k.\text{soc\_compliant}\big) \\ & \wedge\, \text{lithium\_accept}_f\big[\text{lithium\_spec}_k.\text{pi\_code},\, \text{lithium\_spec}_k.\text{section},\, \text{ac\_type}(f)\big] = \text{True}\end{aligned} \tag{14}$$

For non-lithium shipments, $\text{lithium\_ok}(k, f) \equiv \text{True}$ trivially.

Interaction with embargoes. The whitelist acceptance matrix is the *positive* statement of carrier acceptance. The embargo set (Section 5.10) is the *negative* statement of carrier or regulatory refusal, possibly time-bounded. Both must hold: a shipment-arc is lithium-feasible iff acceptance holds AND no active embargo matches. Both predicates are AND-ed in subgraph construction (Section 6).

Aircraft-type dependency. Acceptance differs between freighter and passenger belly for the same carrier and route. The matrix must retain at minimum this binary aircraft-type distinction even if other aircraft-type detail is dropped from the model (parameter $\text{ac\_type}(f)$ is the minimum surviving granularity).

Scope decisions for MVP.

- All six PIs and all three sections supported.
- SOC compliance is shipper-certified boolean; verification deferred to P1.
- UN 38.3 test summary validation deferred to P1 (assumed valid in MVP).
- Per-flight aggregate lithium quantity caps (e.g., "max 50 kg PI965 II per flight") deferred to P1; binary shipment-level feasibility only in MVP.
- PI965 Section IA double-packaging requirements handled at CFS layer, not at routing.
- DDR (damaged / defective / recalled) batteries universally refused; specialized DDR charter routing deferred to P1.
- Out of scope entirely: battery recycling shipments (PI909), vehicle batteries (PI952), emerging chemistries (sodium-ion, solid-state — no standardized PI as of 2026).

Sourcing. Each carrier publishes per-PI acceptance rules as an addendum to the IATA DGR. For MVP, lithium_accept_f is a manually-curated per-carrier table seeded from carrier cargo portals and IATA DGR section 9 (state and operator variations). Annual refresh plus event-driven updates after carrier policy changes (typically following lithium fire incidents).

6 Commodity-Specific Subgraph Construction

Before the MILP is built, we construct $G(N_k, A_k)$ for each commodity k. The subgraph contains only arcs that are time-feasible, cargo-type-compatible, and reachable on a complete $o(k) \to d(k)$ path.

Construction algorithm.

1. **Pickup pass.** For each origin CFS $j \in N_{\text{CFS,O}}$, compute earliest CFS arrival:

$$\tau_k(j) \triangleq t_k^{\text{rdy}} + \mu_{o(k),j}^{\text{PU}}$$

   Arc $(o(k), j) \in A_{\text{PU}}$ is candidate iff there exists a downstream feasible path.

2. **Origin CFS pass.** For each CFS-to-airport arc $(j, a) \in A_{\text{OC}}$ where $j \in N_{\text{CFS,O}}$ and $a \in N_{\text{AP,O}}$, compute earliest airport arrival:

$$\tau_k(a) \triangleq \tau_k(j) + \mu_{j,a}^{\text{OC}} + \beta^{\text{O}}$$

3. **Flight reachability pass.** For each flight $f \in F$ (arc $(a, b) \in A_{\text{AIR}}$): f is candidate for k iff

$$\begin{aligned}
&\tau_k(a) + \text{prep_time}_k \leq \text{CO}_f^* && \text{(meets most restrictive cutoff)} \\
&\text{ETD}_f + \mu_f^{\text{AIR}} \leq T_k^{\text{dead}} - \text{(downstream legs)} && \text{(deadline reachable)} \\
&\text{embargo_ok}(k, f) = \text{True} && \text{(no active embargo matches)} \\
&\text{lithium_ok}(k, f) = \text{True} && \text{(lithium acceptance + SOC + non-DDR)} \\
&\tau_k = \text{General or dgr}(f) = 1 \text{ if } \tau_k = \text{DGR} && \text{(cargo-type compatibility)} \\
&\exists\, u \in U_f : D_k \text{ fits } u && \text{(physical fit in some ULD)}
\end{aligned}$$

   where CO_f^* is the effective cutoff (Eq. 6), embargo_ok(k, f) is the embargo-feasibility predicate (Eq. 12), and lithium_ok(k, f) is the lithium feasibility predicate (Eq. 14). For non-lithium shipments, lithium_ok is trivially True.

4. **Hub transitions pass.** For each pair of consecutive air arcs (f, g) at hub h: compute the *optimistic* arrival time $\tau_k(g_{\text{start}}) = \text{ETA}_f + \min_{u \in U_f \cap U_g} \text{MCT}^*_{f,g,u}$, where $\text{MCT}^*_{f,g,u}$ is defined in (Eq. 29). Optimistic because at subgraph time the ULD type to be used by k is not yet known; pre-filter is conservative on arc admission. P.14 will enforce the exact MCT at solve time based on the actual ULD type selected.

5. **Destination CFS & delivery pass.** For each destination airport $b \in N_{\text{AP,D}}$ reachable: include arc $(b, c) \in A_{\text{DC}}$ where $c \in N_{\text{CFS,D}}$, then $(c, d(k)) \in A_{\text{FD}}$.

6. **Reachability sweep.** Forward BFS from $o(k)$ and backward BFS from $d(k)$. Retain only arcs on paths reachable from both ends.

7 Decision Variables

Variable	Type	Description	Domain
x_{ij}^k	Binary	1 if shipment k routed on arc $(i, j) \in A^k$	$\{0, 1\}$
$y_{f,u,k}^c$	Binary	1 if shipment k uses one ULD of type u on flight f under contract c	$\{0, 1\}$
$z_{f,u}^c$	Integer	Number of ULDs of type u claimed on flight f under contract c	$\mathbb{Z}_{\geq 0}$
$b_{f,k}$	Binary	1 if shipment k tendered to flight f at spot rate (not via contract)	$\{0, 1\}$
$C_{f,u,c}$	Continuous	Effective chargeable weight billed for contract c, flight f, ULD type u (max of actual and pivot)	$\mathbb{R}_{\geq 0}$
$t_k(i)$	Continuous	Arrival time of shipment k at node $i \in N_k$	$\mathbb{R}_{\geq 0}$

8 Constraints

P.1 — Flow Conservation

For each commodity k and each node $n \in N_k$:

$$\sum_{j:(n,j) \in A^k} x_{nj}^k - \sum_{i:(i,n) \in A^k} x_{in}^k = \begin{cases} +1 & \text{if } n = o(k) \\ -1 & \text{if } n = d(k) \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

P.2 — ULD Volume Capacity (per flight, per ULD type, per contract)

For each flight $f \in F$, each ULD type $u \in U_f$, each contract $c \in C_f$:

$$\sum_{k \in K} v_k \cdot y^c_{f,u,k} \;\leq\; \eta \cdot V_u \cdot z^c_{f,u} \tag{16}$$

Total volume loaded into contract-c ULDs of type u on flight f cannot exceed the fill-rate-capped capacity of the claimed ULDs.

P.3 — ULD Weight Capacity

For each f, u, c as above:

$$\sum_{k \in K} cw_k \cdot y^c_{f,u,k} \;\leq\; W_u \cdot z^c_{f,u} \tag{17}$$

P.4 — Per-Flight Contract Allocation Cap

For each flight $f \in F$, ULD type u, contract $c \in C_f$:

$$z^c_{f,u} \;\leq\; \text{alloc}(c, f, u) \tag{18}$$

Forwarder cannot claim more ULDs of type u on flight f than the contract allocates.

P.5 — Aircraft Physical Position Cap (across all contracts)

For each flight $f \in F$, ULD type $u \in U_f$:

$$\sum_{c \in C_f} z^c_{f,u} \;\leq\; P_{f,u} \tag{19}$$

Total ULDs of type u across all contracts cannot exceed the aircraft's physical positions.

P.6 — Period Contract Allocation Cap

For each contract c, ULD type u, period t:

$$\sum_{f \in F_c \cap t} z^c_{f,u} \;\leq\; \text{rem}(c, u, t) \tag{20}$$

Cumulative ULDs claimed in period t under contract c cannot exceed the period's remaining allocation (after prior bookings).

P.7 — Period Minimum Utilization (hard BSA take-or-pay)

For each contract c with $\text{hard}_c = 1$, ULD type u, period t:

$$\sum_{f \in F_c \cap t} z^c_{f,u} \;\geq\; \underline{M}_{c,u,t} \tag{21}$$

$\underline{M}_{c,u,t}$ is the per-run minimum allocation passed in from the upstream period-commitment model. For soft BSAs, $\underline{M}_{c,u,t} = 0$.

P.8 — Arc-to-ULD Linkage

For each shipment k, flight f with $(i,j) = f^{-1}$ in A^k_{AIR}:

$$x^k_{ij} \;=\; \sum_{c \in C_f} \sum_{u \in U_f} y^c_{f,u,k} \;+\; b_{f,k} \tag{22}$$

If shipment k uses arc (i,j), it is loaded via exactly one contract ULD or the spot rate.

P.9 — Spot vs. Contract Exclusivity (per shipment per flight)

Implicit in P.8: at most one of $\{y^c_{f,u,k}\}_{c,u} \cup \{b_{f,k}\}$ equals 1 when $x^k_{ij} = 1$. Formally:

$$\sum_{c \in C_f} \sum_{u \in U_f} y^c_{f,u,k} \;+\; b_{f,k} \;\leq\; 1 \tag{23}$$

P.10 — Effective Chargeable Weight (pivot weight linearization)

For each contract c, flight f, ULD type u — linearizes $\max(\text{actual}, \pi \cdot z)$:

$$C_{f,u,c} \;\geq\; \sum_{k \in K} cw_k \cdot y^c_{f,u,k} \qquad \text{(at least the actual loaded weight)} \tag{24}$$

$$C_{f,u,c} \;\geq\; \pi_{c,u} \cdot z^c_{f,u} \qquad \text{(at least the pivot floor)} \tag{25}$$

Since $C_{f,u,c}$ is minimized in the objective (it multiplies the rate r_c), at optimum $C_{f,u,c} = \max(\sum_k cw_k \cdot y, \pi_{c,u} \cdot z)$ as required. See Section 10 for the linearization argument.

P.11 — Effective Cutoff (Cargo + Documents + Customs Filing)

For each shipment k and each air arc $(i,j) \in A^k_{\text{AIR}}$ with $f = f(i,j)$:

$$t_k(i) + \text{prep\_time}_k \;\leq\; \text{CO}^*_f + M(1 - x^k_{ij}) \tag{26}$$

where CO^*_f is the effective cutoff (Eq. 6) and M is a big-M constant (tight bound: $T^{\text{dead}}_k - t^{\text{rdy}}_k + \epsilon$). The single inequality enforces cargo cutoff, document cutoff, and applicable customs pre-load filing deadlines simultaneously by binding against the most restrictive cutoff in the set.

P.12 — Arrival Time Propagation (ground arcs)

For each shipment k and each ground arc $(i,j) \in A^k_{\text{PU}} \cup A^k_{\text{OC}} \cup A^k_{\text{DC}} \cup A^k_{\text{FD}}$:

$$t_k(j) \;\geq\; t_k(i) + \tilde{\mu}^{\text{G}}_{ij} - M(1 - x^k_{ij}) \tag{27}$$

where $\tilde{\mu}^{\text{G}}_{ij} = \mu^{\text{G}}_{ij} + \beta^{\text{O}}$ for A_{OC}, $\mu^{\text{G}}_{ij} + \beta^{\text{D}}$ for A_{DC}, else μ^{G}_{ij}.

P.13 — Arrival Time Propagation (air arcs)

For each shipment k and each air arc $(i,j) \in A^k_{\text{AIR}}$ with $f = f(i,j)$:

$$t_k(j) \;\geq\; \text{ETA}_f - M(1 - x^k_{ij}) \tag{28}$$

P.14 — Hub Minimum Connect Time

For each shipment k and each pair of consecutive air arcs (f, g) at hub h, the minimum dwell depends on the ULD type used by k on both legs. Define the per-tuple effective MCT (a constant since ψ is a parameter):

$$\mathrm{MCT}^*_{f,g,u} \triangleq \psi_{f,g,u}\,\mathrm{MCT}^{\mathrm{thru}}_{f,g} + (1 - \psi_{f,g,u})\,\mathrm{MCT}^{\mathrm{reULD}}_{f,g} \qquad \forall\, u \in U_f \cap U_g \tag{29}$$

Case 1 — same ULD type on both arcs. For each $u \in U_f \cap U_g$ and each (k, f, g):

$$t_k(g_{\mathrm{start}}) \geq \mathrm{ETA}_f + \mathrm{MCT}^*_{f,g,u} - M\Big(2 - \sum_c y^c_{f,u,k} - \sum_{c'} y^{c'}_{g,u,k}\Big) \tag{30}$$

The big-M term deactivates the constraint unless shipment k uses ULD type u on *both* arcs ($\sum_c y^c_{f,u,k} = 1$ and $\sum_{c'} y^{c'}_{g,u,k} = 1$).

Case 2 — cross-ULD (different type on f vs. g). When k uses different ULD types on f and g (e.g., LD3 on a belly leg, PMC on a freighter leg), breakdown is operationally forced and $\mathrm{MCT}^{\mathrm{reULD}}_{f,g}$ binds:

$$t_k(g_{\mathrm{start}}) \geq \mathrm{ETA}_f + \mathrm{MCT}^{\mathrm{reULD}}_{f,g} - M\Big(2 - x^k_f - x^k_g + \sum_{u \in U_f \cap U_g} \zeta_{f,g,u,k}\Big) \tag{31}$$

where $\zeta_{f,g,u,k}$ is the same-ULD indicator (1 iff k uses u on both arcs), reused from the rehandling-cost linearization (Section 10). The big-M term deactivates this constraint when (i) k does not use both arcs, or (ii) k uses some u on both arcs (Case 1 governs instead).

Joint behavior. ζ and Case 2's deactivation indicator are mutually exclusive within a given (k, f, g) tuple; their sum equals $x^k_f \cdot x^k_g$. Exactly one of (P.14a, P.14b) binds when both arcs are used, and neither binds otherwise.

P.15 — Deadline

For each shipment k:

$$t_k(d(k)) \leq T^{\mathrm{dead}}_k \tag{32}$$

P.16 — Cargo Type Compatibility

For each shipment k with $\tau_k = \mathrm{DGR}$ and each air arc $(i, j) \in A^k_{\mathrm{AIR}}$:

$$x^k_{ij} \leq \mathrm{dgr}(f(i, j)) \tag{33}$$

Analogous constraints for perishable, valuable, heavy cargo.

P.17 — ULD Physical Fit

For each shipment k and each (f, u, c) combination where D_k does not fit ULD type u:

$$y^c_{f,u,k} = 0 \tag{34}$$

This is a pre-processing exclusion, not a runtime constraint — fit is determined from D_k vs. ULD contour at subgraph construction. Bin-packing of multiple shipments inside a single ULD is approximated by the volume constraint P.2; full 3D bin-packing is deferred.

P.18 — Budget Cap (optional)

For each shipment k with $B_k < \infty$:

$$\text{cost}_k \;\leq\; B_k \tag{35}$$

where cost_k is the per-shipment cost contribution from the objective.

P.19 — Domain

$$x_{ij}^k, y_{f,u,k}^c, b_{f,k} \in \{0,1\}, \quad z_{f,u}^c \in \mathbb{Z}_{\geq 0}, \quad C_{f,u,c}, t_k(i) \in \mathbb{R}_{\geq 0} \tag{36}$$

9 Objective Function

Minimize total cost across all shipments:

$$\begin{aligned}
\min \quad & \sum_{k\in K} \sum_{(i,j)\in A_{\text{PU}}^k \cup A_{\text{OC}}^k \cup A_{\text{DC}}^k \cup A_{\text{FD}}^k} c_{ij}^{\text{G}} \cdot x_{ij}^k && \text{(ground cost)} \\
& + \sum_{c\in C} \sum_{f\in F_c} \sum_{u\in U_f} r_c \cdot C_{f,u,c} && \text{(contracted ULD cost incl. pivot floor)} \\
& + \sum_{k\in K} \sum_{f\in F^k} \text{cost}^{\text{spot}}(k,f) \cdot b_{f,k} && \text{(spot rate cost)} \\
& + \sum_{k\in K} \sum_{f\in F^k} \big(\text{FSC}_f + \text{SSC}_f\big) \cdot cw_k \cdot \mathbb{1}[k \text{ on } f] \;+\; (\text{THC, AMS — per shipment per arc}) && \\
& && \text{(surcharges)} \\
& + \sum_{k\in K} \sum_{(f,g)\in \text{Hub}(k)} \sum_{u\in U_f\cap U_g} \rho_{f,g,u}^{\text{reULD}} \cdot (1-\psi_{f,g,u}) \cdot \zeta_{f,g,u,k} && \\
& && \text{(rehandling cost — same-ULD case)} \\
& + \sum_{k\in K} \sum_{(f,g)\in \text{Hub}(k)} \bar{\rho}_{f,g}^{\text{reULD}} \cdot \chi_{f,g,k} && \text{(rehandling cost — cross-ULD case)}
\end{aligned}$$

The same-ULD term uses $\zeta_{f,g,u,k}$ (defined in Section 10): the indicator that shipment k uses ULD type u on *both* f and g. The term contributes only when $\psi_{f,g,u} = 0$ — i.e., same ULD type on both legs but no through-ULD eligibility (e.g., interline pair outside any ULD interchange agreement).

The cross-ULD term uses $\chi_{f,g,k} = x_f^k \cdot x_g^k - \sum_u \zeta_{f,g,u,k}$ (also defined in Section 10): the indicator that k uses both arcs but different ULD types. $\bar{\rho}_{f,g}^{\text{reULD}}$ is the cross-type rehandling cost — typically equal to or slightly higher than the same-type cost, since cargo must be re-stuffed into a new ULD shape.

The two cases together cover all re-ULDing situations exactly: same-u with $\psi = 0$, and cross-u (always forced).

Indicator $\mathbb{1}[k$ on $f]$. Equals 1 if shipment k uses flight f; can be substituted as $\sum_{c,u} y_{f,u,k}^c + b_{f,k}$.

10 Linearization Details

10.1 Pivot Weight Linearization (P.10)

The native cost expression per (contract, flight, ULD type) is:

$$\mathrm{cost}_{f,u,c} = r_c \cdot \max\Big(\sum_k cw_k \cdot y^c_{f,u,k},\ \pi_{c,u} \cdot z^c_{f,u}\Big)$$

The max operator is non-linear. Introduce auxiliary variable $C_{f,u,c} \in \mathbb{R}_{\geq 0}$ with two linear inequality constraints (P.10a, P.10b). Replace the cost term with $r_c \cdot C_{f,u,c}$ in the objective. **Why this works:** the objective minimizes the weighted sum that includes $r_c \cdot C_{f,u,c}$, so the optimizer pushes $C_{f,u,c}$ down to the largest binding constraint, i.e., the maximum of the two right-hand sides. Adds one continuous variable and two inequality constraints per (contract, flight, ULD type) triple.

10.2 Bilinear Rehandling Cost Linearization

The hub-pair indicators required by the objective (rehandling cost) and by P.14 are defined here. The previous draft wrote the rehandling product as $y^c_{f,u,k} \cdot y^{c'}_{g,u,k}$ with implicit contract indices c, c'; the cleaner formulation aggregates over contracts, since a shipment uses at most one contract per (flight, ULD type) per (P.9 spot/contract exclusivity).

Same-ULD indicator $\zeta_{f,g,u,k}$. Binary variable: $\zeta = 1$ iff k uses ULD type u on *both* arcs f and g, across any contract on each.

$$\zeta_{f,g,u,k} \ \leq\ \sum_c y^c_{f,u,k} \tag{37}$$

$$\zeta_{f,g,u,k} \ \leq\ \sum_{c'} y^{c'}_{g,u,k} \tag{38}$$

$$\zeta_{f,g,u,k} \ \geq\ \sum_c y^c_{f,u,k} \ +\ \sum_{c'} y^{c'}_{g,u,k} \ -\ 1 \tag{39}$$

$$\zeta_{f,g,u,k} \ \in\ \{0,1\} \tag{40}$$

The upper bounds force $\zeta = 0$ if either side is 0; the lower bound forces $\zeta = 1$ if both sides are 1. Standard McCormick AND linearization.

Both-arcs-used indicator $\xi_{f,g,k}$. Binary variable: $\xi = 1$ iff k uses both f and g (regardless of ULD type).

$$\xi_{f,g,k} \ \leq\ x^k_f \tag{41}$$

$$\xi_{f,g,k} \ \leq\ x^k_g \tag{42}$$

$$\xi_{f,g,k} \ \geq\ x^k_f + x^k_g - 1 \tag{43}$$

$$\xi_{f,g,k} \ \in\ \{0,1\} \tag{44}$$

Cross-ULD indicator $\chi_{f,g,k}$. Binary variable: $\chi = 1$ iff k uses both arcs but a different ULD type on each.

$$\chi_{f,g,k} \ =\ \xi_{f,g,k} \ -\ \sum_{u \in U_f \cap U_g} \zeta_{f,g,u,k} \tag{45}$$

χ is binary because at most one $\zeta_{f,g,u,k}$ can equal 1 (a shipment uses one ULD type per flight under MVP; cf. deferred item 2 neighborhood). χ enters the objective with positive coefficient (cross-ULD rehandling cost), so the solver has no incentive to inflate it.

Variable count per consecutive hub arc-pair per shipment. $|U_f \cap U_g|$ ζ variables, 1 ξ variable, 1 χ variable; $\approx 5|U| + 5$ constraints. Reused by P.14 (same-ULD and cross-ULD MCT cases) and by the objective's two rehandling-cost terms.

10.3 Big-M Tightening

The big-M constants in P.11, P.12, P.13, P.14 can be replaced with the tightest valid bound: e.g., the maximum possible time span across the planning horizon. Loose big-M values weaken the LP relaxation and slow solving. Practical tightening: use $M = T_k^{\text{dead}} - t_k^{\text{rdy}} + \epsilon$ on time-related constraints.

10.4 Spot Rate Per-Shipment Pre-Computation

The IATA weight-break rule (§5.7) is a min over tiers of piecewise linear cost. Since shipment chargeable weight cw_k is a parameter, $\text{cost}^{\text{spot}}(k, f)$ is fully pre-computed before the MILP is built. No linearization needed at solve time.

11 Re-ULDing — Operational Mechanics

This section motivates and defines re-ULDing in operational detail, which is the basis for the through-ULD compatibility parameter ψ and the rehandling cost ρ^{reULD}.

What re-ULDing is. At an intermediate (hub) airport on a multi-hop route, a built ULD arriving on flight f may need to be *broken down* — i.e., physically dismantled, with cargo removed piece-by-piece — and *rebuilt* into a different ULD for the onward flight g. The specific sequence:

1. ULD arrives on inbound flight f at the hub.
2. Ground handler unloads ULD from aircraft into the hub cargo terminal.
3. ULD is moved to the breakdown area.
4. Cargo is removed from the ULD; each AWB's pieces are extracted and tagged.
5. Cargo is held in the hub warehouse pending the onward flight g.
6. For onward flight g: an empty ULD from carrier(g)'s pool is delivered to build-up.
7. Cargo is restuffed into the new ULD, secured with netting, weighed.
8. Built ULD is moved to the aircraft loading area and loaded onto g.

When re-ULDing is required ($\psi = 0$).

- **Interline (different carriers):** carrier(f) $\neq$ carrier(g). Each airline maintains its own ULD pool; ULDs cannot transfer between pools. Cargo must leave airline f's ULD and enter airline g's ULD.
- **ULD type change:** if the inbound aircraft uses LD3s (wide-body belly) and the outbound aircraft uses PMC main-deck pallets (freighter), the ULD type itself must change. Cargo is re-stuffed into the new ULD type.
- **Hub lacks through-cargo facility for type u:** rare at major hubs (HKG, FRA, NRT all have through-cargo handling) but possible at smaller airports.

- **Forwarder elects re-ULD:** e.g., to re-sort cargo destined to different onward flights into different ULDs. Not modeled in MVP (deferred).

When through-ULD applies ($\psi = 1$).

- **Through-flight technical stop:** same flight number, same aircraft. The ULD stays in its aircraft position throughout the intermediate stop. The hub processing is limited to refueling and crew change. Example: 5Y9123 TPE→ANC→JFK, ULD loaded at TPE stays on the aircraft through ANC and continues to JFK.
- **Same-airline through-cargo at hub:** $\text{carrier}(f) = \text{carrier}(g)$, the airline has a through-cargo agreement at the hub, and the ULD type is compatible with both aircraft. ULD is unloaded from f, moved as a complete unit through the hub through-cargo facility, and loaded onto g. Common at major hubs for same-airline connections (e.g., Cathay TPE→HKG→JFK on CX564 + CX880).

Cost and time impact.

- Through-ULD MCT: MCT^{thru} typically 1.5–4 hours depending on hub and aircraft turnaround.
- Re-ULD MCT: $\text{MCT}^{\text{reULD}}$ typically 6–12 hours including breakdown, hub storage, and rebuild for onward flight.
- Rehandling cost: ρ^{reULD} typically \$150–\$500 per ULD, depending on cargo characteristics and hub labor rates.
- Cargo risk: re-ULDing introduces handling-induced damage and piece-loss risk; valuable / fragile cargo prefers through-ULD where possible (modeled implicitly via the cost differential).

12 Open Items and Future Extensions

1. **Period-level commitment optimization (P.7).** MVP takes $\underline{M}_{c,u,t}$ from a separate upstream model. Future: joint optimization of period commitment allocation across the rolling horizon.

2. **Through-ULD as decision variable.** MVP makes ψ a pre-computed parameter. P1: promote to decision variable, allowing forwarder-elected re-ULDing at compatible hubs for cargo re-sorting.

3. **Probabilistic transit time objective.** MVP uses deterministic mean transit and a hard deadline. P1: extend to $\Pr(\text{arrival} \leq T_k^{\text{dead}}) \geq p$.

4. **Full 3D bin-packing within ULD.** MVP uses aggregate volume constraint (P.2) with $\eta = 0.97$ as a fill-rate proxy. P1: 3D bin-packing model for shipments with awkward piece dimensions.

5. **ULD availability at CFS.** MVP assumes empty ULDs are always delivered to CFS before cargo cutoff. P1: model ULD pool availability as a constraint.

6. **CFS dock door capacity.** MVP assumes infinite CFS throughput. P1: model dock door and labor capacity as a CFS-level constraint.

7. **Cargo type segregation within ULD.** MVP allows any compatible commodities to share a ULD. P1: hazmat compatibility matrix, perishable temperature segregation.

8. **Multiple ULD types per shipment.** MVP forces a shipment to one ULD type per flight. P1: allow splitting across LD3 + LD7 if economically preferred.

9. **Carrier alliance slot sharing.** MVP treats each contract independently. P1: model SkyTeam / Star Alliance / oneworld slot exchanges where ULD positions on partner carriers can substitute.

10. **Spot capacity availability uncertainty.** MVP treats spot rate as always available. Real spot capacity has day-of-departure availability risk. P1: model spot capacity as a stochastic constraint or with explicit booking rejection events.